# Group I, Row V, part 2b: UV term with the phase $e^{-ik\cdot(z-x)p^+/k^+}$ kept

## 1 Answer

Yes, you can keep the phase in the UV term. The phase equals 1 at $x = z$, so the pole is the same as before, and the sum UV + finite is still your 2b exactly. What changes is the finite part of the UV term: the scale becomes $\frac{p^+}{k^+}|\mathbf{k}|$ instead of $1/|y-z|$. Two conditions apply.

1. **Which term.** The divergent term is not just the terms of your bracket that visibly contain $\frac{1}{(x-z)^2}$.
   - Those terms come with $(y-x)^2-(y-z)^2 = O(|x-z|)$, so on their own they behave like $\frac{1}{|x-z|}$.
   - The $\frac{1}{(x-z)^2}$ (in momentum space: the part that falls like $\frac{1}{\mathbf{p}^2}$ at large transverse loop momentum) is the leading term of all six brackets at $x \to z$, averaged over directions. Its coefficient is $\frac{d_\perp}{d}\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$, with $\xi = \frac{p^+}{k^+}$ (`Part2b_UV_finite.pdf`, Step 2).
   - In that term the energy denominator and the colour factor are taken at $x = z$, i.e. $\frac{1}{k^+(y-z)^2}$ and $N_c[U(y)-U(z)]$. Their $x$ dependence is $O(|x-z|)$ and stays in the finite term.
   - The phase is kept.
2. **The direction-dependent leading terms** ($\frac{r^i}{r^2}$, $\frac{r^i(Y\cdot r)}{r^4}-\frac{Y^i}{d\,r^2}$) give no pole. With the phase they no longer integrate to zero, but they are finite. They stay in the finite term.

With the phase, the plain kernel $\frac{1}{(x-z)^2}$ can be used. The phase makes the large-$|x-z|$ region converge for $p^+ \geq \Lambda > 0$, so no cut at $|y-z|$ is needed.

## 2 The UV term with the phase

$$\begin{aligned}\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{1}{k^+}\Big[\frac{d_\perp}{d}\,\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\Big]\\
&\quad\times\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_cU^{af}(y)-N_cU^{af}(z)\Big]\rho^{b'}(x')\rho^f(y)\\
&\quad\times\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\frac{\epsilon}{2}}\int d^dx\; e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{c_d^2}{[(x-z)^2]^{d-1}},\qquad c_d=\Gamma\Big(\frac{d}{2}\Big)\pi^{1-\frac{d}{2}}.\end{aligned}\tag{2.1}$$

**The $x$ integral ($d = 2-\epsilon$).** It is the Fourier transform of a power, with $\mathbf{q} = \frac{p^+}{k^+}\mathbf{k}$:

$$\int d^dr\, e^{i\mathbf{q}\cdot\mathbf{r}}(r^2)^{-a} = \pi^{\frac{d}{2}}2^{d-2a}\frac{\Gamma(\frac{d}{2}-a)}{\Gamma(a)}(q^2)^{a-\frac{d}{2}},\qquad a=d-1.\tag{2.2}$$

This gives

$$\begin{aligned}\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\frac{\epsilon}{2}}\int d^dx\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{c_d^2}{[(x-z)^2]^{d-1}} &= \pi\,\Gamma\Big(\frac{\epsilon}{2}\Big)\frac{\Gamma(1-\frac{\epsilon}{2})^2}{\Gamma(1-\epsilon)}\Big(\frac{\mu^2e^{\gamma_E}}{q^2}\Big)^{\frac{\epsilon}{2}}\\
&= \pi\Big[\frac{2}{\epsilon}-\log\frac{p^{+2}\,\mathbf{k}^2}{k^{+2}\,\mu^2_{\overline{MS}}}\Big]+O(\epsilon)\end{aligned}\tag{2.3}$$

(`sympy`: difference 0). The pole is UV ($\epsilon > 0$). With $\frac{d_\perp}{d}$ in front, $\frac{d_\perp}{d}\pi[\frac{2}{\epsilon}-\ldots] = \frac{d_\perp}{2}\pi[\frac{2}{\epsilon}-\cdots+1]+O(\epsilon)$.

**After the $p^+$ integral (exact).** With $\lambda = \frac{\Lambda}{k^+}$, $\int \frac{dp^+}{k^+} = \int d\xi$ and $\log \xi^2 = 2\log\xi$:

$$\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\rho^f(y)$$
$$\times\Bigg\{\Big[2\log\frac{1-\lambda}{\lambda}-2+4\lambda\Big]\Big[\frac{2}{\epsilon}-\log\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\Big]-2A(\lambda)$$
$$+\frac{d_\perp}{2}\Big(\Big[\frac{1}{6}-\lambda^2+\frac{2}{3}\lambda^3\Big]\Big[\frac{2}{\epsilon}-\log\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}+1\Big]-2B(\lambda)\Big)\Bigg\}, \tag{2.4}$$

with $2\log\frac{1-\lambda}{\lambda} = 2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log(1-\frac{\Lambda}{k^+})$ and

$$A(\lambda) \equiv \int_\lambda^{1-\lambda} d\xi\Big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\log\xi$$
$$= \mathrm{Li}_2(\lambda) - \mathrm{Li}_2(1-\lambda) + \frac{1}{2}\log^2(1-\lambda) - \frac{1}{2}\log^2\lambda - 2(1-\lambda)\log(1-\lambda) + 2\lambda\log\lambda + 2(1-2\lambda),$$
$$B(\lambda) \equiv \int_\lambda^{1-\lambda} d\xi\,\xi(1-\xi)\log\xi = \Big[\frac{\xi^2}{2}\log\xi - \frac{\xi^2}{4} - \frac{\xi^3}{3}\log\xi + \frac{\xi^3}{9}\Big]_\lambda^{1-\lambda} \tag{2.5}$$

(numerically, rel. $2\times10^{-10}$). For $\lambda\to0$: $-2A \to \log^2\frac{\Lambda}{k^+} - 4 + \frac{\pi^2}{3}$ and $-2B \to \frac{5}{18}$ (at $\lambda = 10^{-4}$ the differences are $1.8\times10^{-3}$ and $-9.7\times10^{-8}$).

- $d_\perp = 2-\epsilon$**:** the braces become $\big[2\log\frac{1-\lambda}{\lambda} - \frac{11}{6} + 4\lambda - \lambda^2 + \frac{2}{3}\lambda^3\big]\big[\frac{2}{\epsilon} - \log\frac{\mathbf{k}^2}{\mu^2}\big] - 2A(\lambda) - 2B(\lambda)$.
- $\overline{\rm MS}$**:** drop $\frac{2}{\epsilon}$.
- The logarithm no longer depends on positions, so this UV term is LO (charge at $y$, gluon at $z$) times $\frac{\alpha_s N_c}{2\pi}\{\dots\}$.
- The price is the double logarithm $\log^2\frac{\Lambda}{k^+}$ in $-2A$. It comes from small $\xi$, where the phase cuts the $x$ integral off only at $|x-z| \sim \frac{k^+}{p^+|\mathbf{k}|}$. The finite term below contains the opposite double logarithm.

## 3 The finite term

It is your 2b minus the integrand of Eq. (2.1) at $d=2$. Inside the same $\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\{\dots\}$ and prefactor $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}$, the subtraction is

$$-\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{k^+}\Big[\frac{d_\perp}{2}\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\Big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{1}{(x-z)^2}$$
$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_cU^{af}(y)-N_cU^{af}(z)\Big]\rho^{b'}(x')\rho^f(y). \tag{3.1}$$

- **UV finite.** Numerically, $r^2\langle 2b - \mathrm{CT}\rangle$ falls from $1.5\times10^{-4}$ at $r=10^{-2}$ to $1.5\times10^{-6}$ at $r=10^{-3}$ ($d_\perp = 2,3$; $\xi = 0.02, 0.37, 0.95$; phase kept in both).
- **Large $|x-z|$.** Both terms behave like $\frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{r^2}$, which converges by oscillation. For example $2\pi\int_R^{10R} dr\frac{J_0(qr)}{r}$, for $R = 10, 100, 1000$, is 9.889, −0.889, −0.032 at $\xi = 0.02$ and 0.292, 0.010, $7.1\times10^{-5}$ at $\xi = 0.37$.

## 4 Relation to the split without the phase

Per $\xi$, the two UV kernels differ by a finite amount (2 dims, $\mathbf{q} = \xi\mathbf{k}$, $Y = y-z$):

$$\int d^2x\Big[\frac{e^{i\mathbf{q}\cdot(x-z)}}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\Big] = -\pi\log\frac{\mathbf{q}^2Y^2e^{2\gamma_E}}{4} \tag{4.1}$$

(numerically: $q=1.3$, $Y=0.7$: 1.320991 vs. 1.320991; $q=0.2$, $Y=2.0$: 6.485644 vs. 6.485644; $q=3.0$, $Y=0.4$: −0.417140 vs. −0.417141). Indeed $\pi[\frac{2}{\epsilon} - \log\frac{q^2}{\mu^2}] - \pi[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{Y^2\mu^2}{4}]$ is exactly this. So:

- the poles of the two UV terms agree;
- the UV terms differ by a finite term;
- the finite terms differ by the opposite amount;
- the total is unchanged.

## Checks

Script `part2b_phase_checks.py` → `part2b_phase_checks.json`. The short-distance coefficient with $d_\perp \neq d$ is checked in `part2b_dperp_checks.py`.